ANSHU SINGH

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EDUCATION

Vellore Institute of Technology **Sep. 2023 - May 2027**
Bachelor of Technology in Computer Science *CGPA: 8.71/10*

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, HTML/CSS, SQL
Frameworks & Libraries: FastAPI, Next.js, React, Node.js, Express.js, Tailwind CSS, Prisma ORM, PyTorch
AI, Machine Learning & Data: Large Language Models (LLMs), AI Agents (ReAct), Multi-Agent Systems, RAG, Graph DBs, Vector Search, Transformers, OCR, scikit-learn, Pandas, NumPy, Pydantic
Databases, Cloud & Tools: PostgreSQL, SQLite, Supabase, KuzuDB, ChromaDB, REST APIs, WebSockets, Git, GitHub, Docker, Render, AWS

PROJECTS

EPC Data Centre Intelligence Platform | *FastAPI, Next.js, KuzuDB, ChromaDB, Gemini Vision, React* | Link
- Built a 5-agent full-stack platform coordinating engineering specs, procurement, and Level 1-5 commissioning across 15,000+ data centre equipment items.
- Coupled KuzuDB graph queries with ChromaDB vector search to compute Critical Path Method (CPM) risks and reroute suppliers, cutting lead times from weeks to days.

ClinicalAI - Discharge Summary Agent | *Python, FastAPI, OCR, LLM APIs, Pydantic, PyTorch* | Link
- Engineered an autonomous clinical discharge agent parsing scanned medical records via OCR, running a bounded 10-step ReAct loop across 4 diagnostic audit tools.
- Enforced zero-fabrication Pydantic schemas and multi-provider LLM failover, reaching 0.0 normalized edit distance by iteration 2 in clinician feedback evaluations.

Factualize - PDF Fact-Checking Engine | *Python, FastAPI, Gemini API, Docling, SQLite, React* | Link
- Architected a local document intelligence tool that extracts verified claims and detects cross-document contradictions using FastAPI, PyMuPDF, and Docling OCR.
- Integrated Google Gemini with Pydantic schemas, tracking daily rate ceilings in SQLite and linking conclusions back to exact source text with 100% auditability.

Society Maintenance Tracker | *Next.js, TypeScript, PostgreSQL, Prisma, Tailwind CSS* | Link
- Built a full-stack apartment ticketing portal with Next.js and Prisma, giving residents request tracking and administrators a triage queue across 5 categories.
- Implemented NextAuth JWT sessions, Supabase image storage up to 5MB, Resend email alerts, status audit logs, and an overdue ticket engine set from 1 to 90 days.

OPEN-SOURCE CONTRIBUTIONS

- **Hugging Face Transformers -** Fixed inverse_sqrt learning rate scheduling so optimizer-specific kwargs, including timescale, are forwarded correctly in PyTorch training runs.
- **Prime Intellect AI Community Environments -** Added an MMMU-Pro multimodal-reasoning benchmark environment compatible with RL verifiers, implementing data loaders, reward logic, and eval metrics.
- **Future AGI -** Built three React frontend features: clickable error drill-downs, sortable dashboard headers with formatted dates, and validated task cloning, covered by Jest unit tests.

CERTIFICATIONS

- **Amazon Machine Learning Summer School – Amazon, 2026**
- **Applied Machine Learning in Python** - University of Michigan (Coursera)
- **AWS Technical Essentials -** Amazon Web Services
- **Cloud Computing** - NPTEL / IIT Kharagpur (75%), 2025

EXTRACURRICULAR

Basketball Team **Jul. 2024 - Aug. 2024**
Captain *Vellore Institute of Technology*
- Captained a 6-player intra-college basketball squad, demonstrating leadership and decision-making under competitive pressure
- Mentored teammates as team Captain, organizing practices and fostering communication and collaboration across the squad.